
(a) Perception
Monocular video
Event detection
Keypoint detection
and needle segmentation
reposition end (t = 0.0 s)
entry (t = 4.6 s)
t = 0.7 s
t = 1.3 s
t = 2.5 s
t = 3.4 s
t = 4.6 s
needle mask
needle tip
grasp point
jaw tip / jaw base
(b) Measurement and scoring
Needle Hold Ratio
ratio
EASE
1.0
0.75
0.5
0.0
poor
good
poor
t
0
2
4
Needle Hold Angle
angle
EASE
180°
150°
135°
90°
0°
1
2
3
1
t
Depth of Needle Hold
depth
EASE
1.0
0.5
0.25
0.0
1
2
3
t